

KAITAIA, New Zealand

WMO: 930120

Lat: 35.1328S

Lon: 173.2639E

Elev: 85

StdP: 100.31

Time Zone: 12.00 (NZL)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	6.3	2.6	4.6	7.8	3.6	5.0	8.5	15.4	13.5	14.4	13.3	2.4	100	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	24.9	19.2	24.0	18.9	23.1	18.5	21.1	22.5	20.4	22.1	19.8	21.7	5.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.5	21.8	19.9	14.8	21.1	19.2	14.1	20.6	61.8	22.4	59.3	22.1	57.1	21.6	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.1	9.9	DB	3.3	27.1	0.9	1.2	2.7	27.9	2.2	28.6	1.7	29.2	1.0	30.0	(4)
(5)			WB	3.0	22.0	0.9	1.0	2.4	22.7	1.9	23.3	1.4	23.9	0.8	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	19.2	19.4	18.1	16.0	14.1	12.5	11.7	12.0	12.7	14.1	15.3	17.4	(6)
(7)		DBStd	3.15	1.62	1.54	1.83	1.87	1.82	1.86	1.69	1.54	1.68	1.52	1.53	1.69	(7)
(8)		HDD10.0	9	0	0	0	0	0	2	4	2	1	0	0	0	(8)
(9)		HDD18.3	1256	9	7	26	72	131	176	206	196	171	130	92	39	(9)
(10)		CDD10.0	1899	285	262	251	181	128	76	57	64	81	128	158	229	(10)
(11)		CDD18.3	103	36	36	18	2	1	0	0	0	0	0	1	10	(11)
(12)		CDH23.3	142	56	66	16	0	0	0	0	0	0	0	0	5	(12)
(13)		CDH26.7	1	0	1	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.4	4.0	3.9	3.8	4.4	3.9	4.5	4.7	5.0	5.1	4.7	4.2	(14)
(15)	Precipitation	PrecAvg	1338	75	94	82	107	124	158	154	146	122	98	100	87	(15)
(16)		PrecMax	1855	287	373	226	329	324	352	358	309	234	213	221	185	(16)
(17)		PrecMin	981	2	7	2	10	29	57	25	40	40	11	32	18	(17)
(18)		PrecStd	242	59	82	53	62	61	63	65	59	44	49	47	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.9	26.2	24.9	22.0	19.9	18.1	17.0	18.6	20.0	21.6	24.3	(19)	
(20)			MCWB	19.7	19.4	19.8	17.1	17.5	17.0	14.9	14.3	15.9	15.7	17.0	18.4	(20)
(21)		2%	DB	24.8	25.1	23.7	21.0	18.9	17.2	15.9	16.1	17.5	18.7	20.5	22.9	(21)
(22)			MCWB	19.5	18.8	18.9	16.7	16.4	15.7	14.1	13.8	15.2	15.3	16.6	18.2	(22)
(23)		5%	DB	24.0	24.0	22.7	20.1	18.2	16.5	15.2	15.5	16.5	17.9	19.7	22.1	(23)
(24)			MCWB	19.3	18.6	18.5	16.6	15.8	14.8	13.3	13.4	14.0	15.0	16.2	17.7	(24)
(25)		10%	DB	23.0	23.1	21.8	19.4	17.4	15.7	14.5	14.9	15.6	17.3	18.9	21.3	(25)
(26)			MCWB	18.8	18.2	18.0	16.4	15.2	13.8	12.7	13.0	13.2	14.6	15.8	17.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	21.8	22.2	19.2	19.1	17.6	15.6	15.6	17.7	17.6	19.3	20.4	(27)
(28)			MCD B	23.8	23.0	23.4	19.9	19.5	17.9	16.2	16.2	17.9	18.5	19.8	21.7	(28)
(29)		2%	WB	21.1	20.6	21.2	18.7	17.7	16.5	14.7	14.9	15.7	16.8	18.0	19.8	(29)
(30)			MCD B	22.5	22.7	22.3	19.6	18.2	17.0	15.5	15.6	16.6	17.7	19.2	21.2	(30)
(31)		5%	WB	20.5	19.9	19.7	17.9	16.5	15.4	14.0	14.1	14.7	16.1	17.3	19.2	(31)
(32)			MCD B	22.1	22.5	21.2	19.1	17.5	16.0	14.9	15.0	15.9	17.1	18.6	20.7	(32)
(33)		10%	WB	19.9	19.2	18.7	17.0	15.6	14.4	13.2	13.4	13.9	15.4	16.5	18.5	(33)
(34)			MCD B	21.8	21.8	20.9	18.7	16.9	15.3	14.1	14.5	15.2	16.6	18.1	20.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.4	7.1	6.3	5.8	5.5	5.4	5.3	5.6	5.9	6.1	7.0	(35)
(36)			MCDBR	8.2	8.8	8.0	6.8	6.4	5.7	5.4	5.7	6.3	6.5	7.7	8.0	(36)
(37)			MCWBR	3.8	3.8	3.9	3.8	4.2	4.1	3.8	3.9	3.9	4.2	4.5	4.1	(37)
(38)		5% WB	MCD BR	5.3	6.9	5.3	5.1	4.7	4.3	4.2	4.5	4.6	5.1	5.2	5.6	(38)
(39)			MCWBR	3.1	3.8	3.5	3.7	3.7	3.9	3.5	3.7	3.4	3.9	3.9	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.340	0.328	0.317	0.311	0.308	0.303	0.312	0.335	0.329	0.334	0.340	(40)	
(41)		taud	2.573	2.620	2.645	2.665	2.653	2.641	2.658	2.621	2.528	2.560	2.567	2.599	(41)	
(42)		Ebn at Noon	991	975	951	906	856	830	857	898	927	975	996	1000	(42)	
(43)		Edh at Noon	106	97	88	77	68	65	67	78	97	102	106	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.86	6.06	4.96	3.63	2.58	2.12	2.30	3.15	4.26	5.48	6.28	6.53	(44)	
(45)		RadStd	0.40	0.42	0.31	0.21	0.13	0.12	0.14	0.18	0.21	0.34	0.40	0.52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page